

USB Camera Display

12.1 Camera Reading

`capture=cv.VideoCapture(0)`

Parameter meaning:

The parameter in `VideoCapture()` is 0, which means to open the laptop's built-in camera. If the parameter is a video file path, it opens the video, such as `cap = cv2.VideoCapture("../test.avi")`

12.2 Display Camera Video

`ret,img = frame.read()`

Return value meaning:

`ret`: `ret` is a bool value that indicates whether the correct frame was read back.

`img`: Image data for each frame.

12.3 Code and Actual Effect Display

Code path: `/home/yahboom/Dofbot/6.AI_Visual/0.Camera Driver Tutorial.ipynb`

Main code:

```
import cv2
import ipywidgets.widgets as widgets
import threading
import time

image_widget = widgets.Image(format='jpeg', width=600, height=500) # Set camera
display_component
display(image_widget) # Display
camera_component
```

```
#bgr8 to jpeg format
import enum
import cv2

def bgr8_to_jpeg(value, quality=75):
    return bytes(cv2.imencode('.jpg', value)[1])
```

```
image = cv2.VideoCapture(0,cv2.CAP_V4L2) # Open camera
ret, frame = image.read()
image_widget.value = bgr8_to_jpeg(frame)
```

After running the code block, the USB camera feed will be displayed.

